

Problem Set 2

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Problem 1. Tree's Degrees

(a)

Proof. Let $P(n)$ be the proposition that any tree with $n \geq 2$ vertices and k vertices of degree ≥ 3 has at least $k + 2$ leaves.

Base Case: $n = 2$; then k must be 0, so there are at most $0 + 2$ leaves. The base case holds.

Inductive step: Assume $P(n - 1)$ is true for $n > 2$.

Consider a tree T with $n > 2$ vertices, of which k have degree at least 3. Let $T' = T - l$, where l is a leaf of T . Consider cases based on the leaf count of T' .

- **Case I:** If l 's parent had more than one child, the leaf count decreases by 1. This implies the parent has degree ≥ 3 , so $|L'| = |L| - 1$ and $k' = k - 1$.
- **Case II:** If l 's parent had only one child, the leaf count doesn't change. This implies the parent has degree exactly 2, so $|L'| = |L|$ and $k' = k$.

Now Induction hypothesis on T' we get $|L'| \geq k' + 2$ and now adding vertex l back and substituting value $|L'|$ and k' From case I we get $|L| \geq k + 2$ and case II also results in same equation. Thus, by The Induction principle, $P(n)$ holds for all $n \geq 2$.

□

(b)

Proof. By the property of trees, the total number of edges $|E| = n - 1$. Now, substituting it into Handshake Lemma yields $\sum \deg(v) = 2n - 2$.

Let l be the number of leaves of a tree, also k is vertices of degree at least 3, and let $n - l - k$ be the number of vertices of degree 2. so total degree of a tree,

$$\begin{aligned} \sum \deg(v) &\geq l + 2(n - l - k) + 3k \\ 2n - 2 &\geq l + 2n - 2l - 2k + 3k \\ l &\geq k + 2 \end{aligned}$$

□

Problem 2. Froli-King

(a) By modeling the chessboard as a graph where nodes represent squares of the chessboard, and there exists an edge between two node if they are adjacent. Now a move for the king is legal if two squares are adjacent to each other, and the total number of legal king moves is 420 and In our graph model, every legal move counts each edge twice because moves $a \rightarrow b$ and $b \rightarrow a$ are Considered distinct moves, so the total number of edges in our graph $|E| = 420/2 = 210$

Frolicsome corresponds to an Euler tour because the graph model (a) says we must start and end at the same vertex, and (b) and (c) implies we must visit every edge exactly once. Also, since our graph has exactly 210 edges and a move corresponds to an edge, Frolicsome of 210 moves is just an Euler tour.

And we know an Euler tour is only possible iff every vertex is connected and has even degree. But the corner nodes have degree of 3, which is odd.

(b)

To make our graph Eulerian, we must make every vertex of even degree, and from part(a) there are a total of 28 corner vertices with odd degree, and since they are connected, we can remove one edge between two odd-degree vertices to make them even degree, so the total number of edges we must remove is $28/2 = 14$ and if we remove corner edges alternatively, it will result in a graph with a total of 196 edges; also, each vertex is connected and has even degree.

Corollary: There exists a Frolicsome sequence of 196 King moves on this 8×8 board

Problem 3. DAGs

(a) The largest chain is (1, 2, 3, 6, 9, 10, 12).

(b) The largest anti-chain is (1,4,5,7,8,11) because the remaining nodes form a chain as shown in part (a), and every element of the anti-chain is incomparable to the remaining elements. Therefore, it's the largest possible anti-chain.

(c) 12 second

(d) 7 second

(e) 3 processors.

(e) **Minimum Processors and Schedule**

The smallest number of processors is $p = 3$.

Optimal Schedule:

Time Step	Processor 1	Processor 2	Processor 3
$t = 1$	1	7	8
$t = 2$	2	4	5
$t = 3$	3	11	Idle
$t = 4$	6	Idle	Idle
$t = 5$	9	Idle	Idle
$t = 6$	10	Idle	Idle
$t = 7$	12	Idle	Idle

Proof that 2 processors take longer: For processors to do task 6 on $t = 4$, 7 tasks must be done in 3 seconds with two processors maximum task can be done in 3 seconds $= 3 \times 2 = 6$ which is less than 7. so p-1 or less processors will take longer than Optimal time.

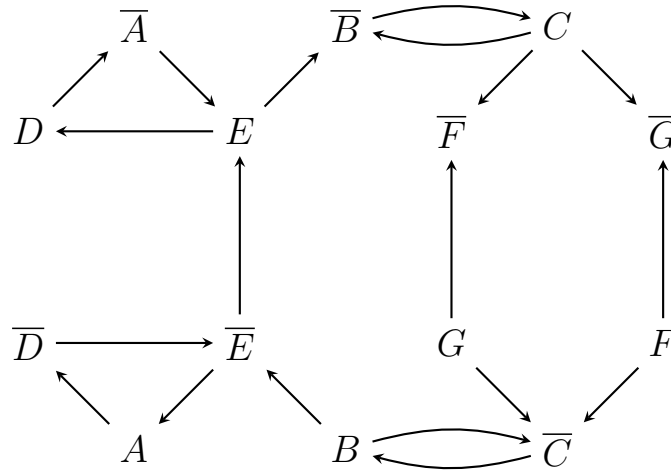
(f) (1, 7, 8, 11, 4, 5, 2, 3, 6, 9, 10, 12)

Problem 4. Pokémon Constraints

(a)

$$\phi_2 := (\bar{A} \vee \bar{D}) (\bar{B} \vee \bar{E}) \wedge (\bar{C} \vee \bar{F}) \wedge (\bar{C} \vee \bar{G}) \wedge (\bar{F} \vee \bar{G})$$

(b)



(c) $P(k)$:= For any arbitrary literals x and y , if ψ is true and implication graph of ψ contains path π of length k from x to y then $x \implies y$.

proof by Induction. Let ψ be 2-CNF formula, and let y, x be literals in ψ . now assume ψ is true and the implication graph of ψ contains a path of length k from x to y .

Base Case ($k = 1$): Path of length 1 from x to y implies there is an edge directly between x and y and by the definition of an edge in the implication graph, $x \implies y$.

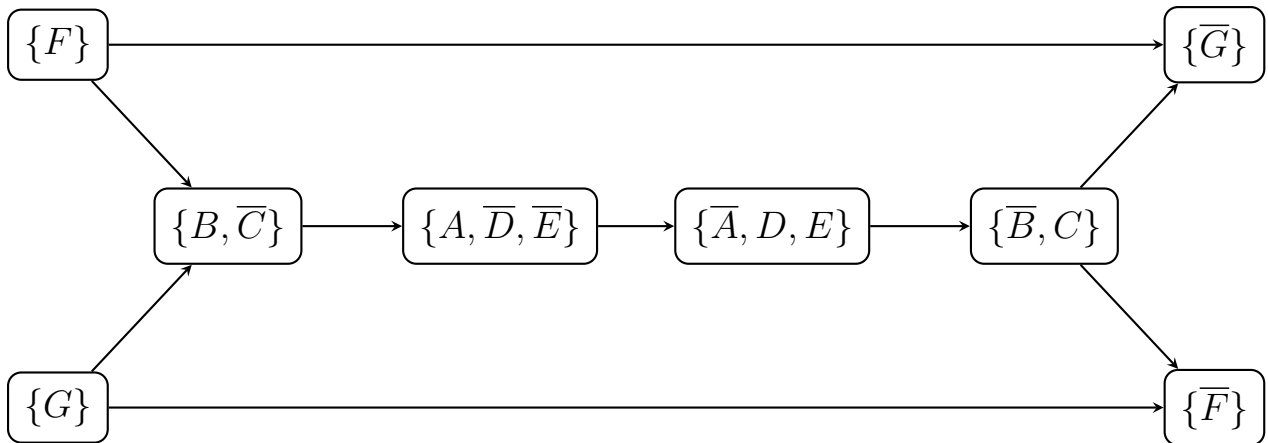
Inductive step: Suppose for $P(k - 1)$ is true for all $k > 1$. Now a path of length k from x to y can be shown as

$$x = v_0 \rightarrow v_1 \rightarrow v_2 \cdots \rightarrow v_{k-1} \rightarrow v_k = y$$

Since the path from x to v_{k-1} is of length $k-1$, by the induction hypothesis we know $x \implies v_{k-1}$, and by the definition of an edge in the given graph, $v_{k-1} \implies y$. Therefore, by the transitive property of implication $x \implies y$.

So by induction hypothesis $P(k)$ is true for all $k > 0$.

(d)



(e) Topological order of SCCs of $H = (\{F\}, \{G\}, \{B, \bar{C}\}, \{A, \bar{D}, \bar{E}\}, \{\bar{A}, D, E\}, \{\bar{B}, C\}, \{\bar{G}\}, \{\bar{F}\})$
 $T = \{\text{Chikorita}, \text{Dratini}, \text{Eevee}\}$ which satisfies all constraints.